

6 Lecture 6

Determinants

Definition 6.1. The determinant of a matrix $A = [a_{i,j}] \in \text{Mat}_n(\mathbb{F})$, denoted $\det(A)$ or $|A|$, is defined as

$$\det(A) \stackrel{\text{def}}{=} \sum_{f \in S_n} \left(\text{sgn}(f) \cdot \prod_{i \in [n]} a_{i,f(i)} \right), \quad (6.1)$$

where the notation $\sum_{f \in S_n}$ means that we sum over all permutations f of S_n (since there are $n!$ elements in S_n , we have $n!$ summands of the form $\text{sgn}(f) \cdot \prod_{i \in [n]} a_{i,f(i)}$ in Formula (6.1)). Expression (6.1) is called the *Leibniz formula*. Note that $\det(A)$ is an element of $\mathbb{F}$, therefore, one may say that the determinant is a function $\det: \text{Mat}_n(\mathbb{F}) \rightarrow \mathbb{F}$ defined by (6.1).

Formula (6.1) may appear daunting, but it is actually fairly simple. Every term of the sum, $\prod_{i \in [n]} a_{i,f(i)}$, is a product of n factors containing one and only one element from each row of A and one and only one element from each column of A . There there are $|S_n| = n!$ such products. Each product has a plus or minus sign attached to it according as the column indices form an even or odd permutation when the row indices are in the natural order (i.e. 1, 2, 3, ...).

For example.

If $n = 2$ then $S_2 = \left\{ id_2 = \begin{pmatrix} 1 & 2 \\ 1 & 2 \end{pmatrix}, f = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \right\}$ and for any $A = \begin{bmatrix} a_{1,1} & a_{1,2} \\ a_{2,1} & a_{2,2} \end{bmatrix} \in \text{Mat}_2(\mathbb{F})$ we have:

$$\begin{aligned} \det(A) &= \text{sgn}(id_2) \cdot a_{1,id_2(1)} \cdot a_{2,id_2(2)} + \text{sgn}(f) \cdot a_{1,f(1)} \cdot a_{2,f(2)} \\ &= a_{1,1}a_{2,2} - a_{1,2}a_{2,1}. \end{aligned}$$

If $n = 3$ then, using the notations from the table on page 24, for any $A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \in \text{Mat}_3(\mathbb{F})$ we have:

$$\begin{aligned} \det(A) &= \text{sgn}(id_3) \cdot a_{1,id_3(1)}a_{2,id_3(2)}a_{3,id_3(3)} + \text{sgn}(f_1) \cdot a_{1,f_1(1)}a_{2,f_1(2)}a_{3,f_1(3)} + \text{sgn}(f_2) \cdot a_{1,f_2(1)}a_{2,f_2(2)}a_{3,f_2(3)} + \\ &\quad + \text{sgn}(f_3) \cdot a_{1,f_3(1)}a_{2,f_3(2)}a_{3,f_3(3)} + \text{sgn}(f_4) \cdot a_{1,f_4(1)}a_{2,f_4(2)}a_{3,f_4(3)} + \text{sgn}(f_5) \cdot a_{1,f_5(1)}a_{2,f_5(2)}a_{3,f_5(3)} \\ &= a_{1,1}a_{2,2}a_{3,3} + a_{1,2}a_{2,3}a_{3,1} - a_{1,3}a_{2,2}a_{3,1} - a_{1,1}a_{2,3}a_{3,2} - a_{1,2}a_{2,1}a_{3,3} + a_{1,3}a_{2,1}a_{3,2}. \end{aligned}$$

Thus, we proofed *Sarrus' rule*: the determinant of a 3-by-3 matrix is the sum of the products along the solid diagonals minus the sum of the products along the dashed diagonals:

$$\begin{array}{ccccc} & + & + & + & \\ & a_{11} & a_{12} & a_{13} & \\ a_{21} & \swarrow & \nearrow & \swarrow & \nearrow \\ & a_{22} & a_{23} & & \\ a_{31} & \cdots & \cdots & \cdots & \\ & - & - & - & \end{array} \quad \begin{array}{ccccc} & a_{11} & a_{12} & & \\ & a_{21} & a_{22} & & \\ & a_{31} & a_{32} & & \end{array}$$

WARNING! Sarrus' rule can be applied *only* to 3-by-3 matrices!

Exercise 1. For any $f, g \in S_n$ and $A = [a_{i,j}] \in \text{Mat}_n(\mathbb{F})$ we have

$$\prod_{i \in [n]} a_{i,f(i)} = \prod_{i \in [n]} a_{g(i),(f \circ g)(i)}. \quad (6.2)$$

[hint: the statement is trivial, at the base level, it means that n factors, $a_{1,f(1)}, \dots, a_{n,f(n)}$ can be rearranged in any way without altering the total product; for example, if $g = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$ and $f \in S_3$, then we have $a_{1,f(1)}a_{2,f(2)}a_{3,f(3)} = a_{3,f(3)}a_{1,f(1)}a_{2,f(2)} = a_{g(1),(f \circ g)(1)}a_{g(2),(f \circ g)(2)}a_{g(3),(f \circ g)(3)} \cdot]$

Statement 6.1. For any $A \in \text{Mat}_n(\mathbb{F})$ we have

$$\det(A^T) = \det(A). \quad (6.3)$$

Proof. Suppose that $A = [a_{i,j}]$ and $A^T = [b_{i,j}]$, then, by Definition 2.1, $b_{i,j} = a_{j,i}$ for all $i, j \in [n]$. We have:

$$\begin{aligned} \det(B) &\stackrel{(6.1)}{=} \sum_{f \in S_n} \left(\text{sgn}(f) \cdot \prod_{i \in [n]} b_{i,f(i)} \right) \\ &\stackrel{(6.2)}{=} \sum_{f \in S_n} \left(\text{sgn}(f) \cdot \prod_{i \in [n]} b_{f^{-1}(i),(f \circ f^{-1})(i)} \right) \\ &= \sum_{f \in S_n} \left(\text{sgn}(f) \cdot \prod_{i \in [n]} b_{f^{-1}(i),i} \right) \\ &= \sum_{f \in S_n} \left(\text{sgn}(f) \cdot \prod_{i \in [n]} a_{i,f^{-1}(i)} \right) \\ &\stackrel{(5.1)}{=} \sum_{f \in S_n} \left(\text{sgn}(f^{-1}) \cdot \prod_{i \in [n]} a_{i,f^{-1}(i)} \right) \\ &\stackrel{(6.1)}{=} \det(A). \end{aligned}$$

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Lemma 6.1. If $f \in S_n$ and $f \neq \text{id}_n$ then there exists $i' \in [n]$ such that $f(i') < i'$.

Proof. Since all elements of S_n are bijections from $[n]$ to $[n]$, for any $f \in S_n$ we have:

$$\sum_{i \in [n]} f(i) = \sum_{i \in [n]} i = \frac{n(n+1)}{2}. \quad (6.4)$$

Suppose that $f \neq \text{id}_n$ and $f(i) \geq i$ for any $i \in [n]$. Since $f \neq \text{id}_n$, this implies that there exists $k \in [n]$ such that $f(k) > k$. But, in this case, the following estimate contradicts (6.4):

$$\begin{aligned} \sum_{i \in [n]} f(i) &= f(k) + \sum_{\substack{i \in [n] \\ i \neq k}} f(i) \\ &\geq f(k) + \sum_{\substack{i \in [n] \\ i \neq k}} i \\ &> k + \sum_{\substack{i \in [n] \\ i \neq k}} i \\ &= \frac{n(n+1)}{2} \end{aligned}$$

■

Proposition 6.1. *[Three Main Properties of Determinants]*

- 1) if $A \in \text{Mat}_n(\mathbb{F})$ and there exist $r, s \in [n]$, $r \neq s$ such that $[A]_{(r)} = [A]_{(s)}$ then $\det(A) = 0$;
- 2) if $A, B, C \in \text{Mat}_n(\mathbb{F})$ and there exists $r \in [n]$ such that $[C]_{(r)} = \lambda_1 [A]_{(r)} + \lambda_2 [B]_{(r)}$ for some $\lambda_1, \lambda_2 \in \mathbb{F}$ and $[A]_{(i)} = [B]_{(i)} = [C]_{(i)}$ for each $i \in [n]$, $i \neq r$, then $\det(C) = \lambda_1 \cdot \det(A) + \lambda_2 \cdot \det(B)$;
- 3) if $A \in \text{Mat}_n(\mathbb{F})$ is a triangular matrix² then $\det(A) = \prod_{i \in [n]} [A](i, i)$.

For example:

Item 1 of Proposition 6.1 tells us that:

$$\det \begin{bmatrix} a & b & c \\ d & f & g \\ a & b & c \end{bmatrix} = 0$$

(since rows 1 and 3 of the matrix are the same);

Item 2 of Proposition 6.1 tells us that:

$$\det \begin{bmatrix} a & b & c \\ d + d' & f + f' & g + g' \\ h & i & j \end{bmatrix} = \det \begin{bmatrix} a & b & c \\ d & f & g \\ h & i & j \end{bmatrix} + \det \begin{bmatrix} a & b & c \\ d' & f' & g' \\ h & i & j \end{bmatrix};$$

Item 3 of Proposition 6.1 tells us that:

$$\det \begin{bmatrix} a & b & c \\ 0 & f & g \\ 0 & 0 & j \end{bmatrix} = a \cdot f \cdot j.$$

(since the matrix is triangular).

Proof. 1) Suppose that $A = [a_{i,j}]$ and let A_n be the subset of all even permutations of S_n ³. We define the sums X_1 and X_2 as follows

$$X_1 = \sum_{f \in A_n} \left(\prod_{i \in [n]} a_{i, f(i)} \right) \quad \text{and} \quad X_2 = \sum_{f \in S_n \setminus A_n} \left(\prod_{i \in [n]} a_{i, f(i)} \right).$$

Since $\text{sgn}(f) = 1$ for any $f \in A_n$ and $\text{sgn}(f) = -1$ for any $f \in S_n \setminus A_n$, from (6.1), we have

$$\det(A) = X_1 - X_2.$$

²See Definition 1.8.

³ A_n is called the *alternating group* of degree n .

Let $\tau \in S_n$ be a transposition such that $\tau(r) = s$ and $\tau(s) = r$. Then we have:

$$\begin{aligned}
X_1 &= \sum_{f \in A_n} \left(a_{r,f(r)} \cdot a_{s,f(s)} \cdot \prod_{\substack{i \in [n] \\ i \neq r,s}} a_{i,f(i)} \right) && // \text{by the definition of } X_1 // \\
&= \sum_{f \in A_n} \left(a_{s,f(r)} \cdot a_{r,f(s)} \cdot \prod_{\substack{i \in [n] \\ i \neq r,s}} a_{i,f(i)} \right) && // \text{since } a_{r,f(r)} = a_{s,f(r)} \text{ and } a_{s,f(s)} = a_{r,f(s)} // \\
&= \sum_{f \in A_n} \left(a_{\tau(s), (f \circ \tau)(r)} \cdot a_{\tau(r), (f \circ \tau)(s)} \cdot \prod_{\substack{i \in [n] \\ i \neq r,s}} a_{\tau(i), (f \circ \tau)(i)} \right) && // \text{by Equality (6.2), for } g = \tau // \\
&= \sum_{f \in A_n} \left(a_{r, (f \circ \tau)(r)} \cdot a_{s, (f \circ \tau)(s)} \cdot \prod_{\substack{i \in [n] \\ i \neq r,s}} a_{i, (f \circ \tau)(i)} \right) && // \text{since } \tau(r) = s, \tau(s) = r \text{ and } \tau(i) = i, i \neq r, s // \\
&= \sum_{f \in S_n \setminus A_n} \left(a_{r,f(r)} \cdot a_{s,f(s)} \cdot \prod_{\substack{i \in [n] \\ i \neq r,s}} a_{i,f(i)} \right) && // \text{since } f \mapsto f \circ \tau \text{ is a bijection from } A_n \text{ to } S_n \setminus A_n // \\
&= X_2. && // \text{by the definition of } X_2 //
\end{aligned}$$

Thus, $\det(A) = X_1 - X_2 = X_2 - X_2 = 0$.

2) Suppose that $A = [a_{i,j}]$, $B = [b_{i,j}]$ and $C = [c_{i,j}]$, then we have:

$$\begin{aligned}
\det(C) &= \sum_{f \in S_n} \left(\text{sgn}(f) \cdot c_{r,f(r)} \cdot \prod_{\substack{i \in [n] \\ i \neq r}} c_{i,f(i)} \right) && // \text{by Equality (6.1)} // \\
&= \sum_{f \in S_n} \left(\text{sgn}(f) \cdot (\lambda_1 \cdot a_{r,f(r)} + \lambda_2 \cdot b_{r,f(r)}) \cdot \prod_{\substack{i \in [n] \\ i \neq r}} c_{i,f(i)} \right) && // \text{since } c_{r,f(r)} = \lambda_1 \cdot a_{r,f(r)} + \lambda_2 \cdot b_{r,f(r)} // \\
&= \lambda_1 \cdot \sum_{f \in S_n} \left(\text{sgn}(f) \cdot a_{r,f(r)} \cdot \prod_{\substack{i \in [n] \\ i \neq r}} c_{i,f(i)} \right) + \\
&\quad + \lambda_2 \cdot \sum_{f \in S_n} \left(\text{sgn}(f) \cdot b_{r,f(r)} \cdot \prod_{\substack{i \in [n] \\ i \neq r}} c_{i,f(i)} \right) \\
&= \lambda_1 \cdot \sum_{f \in S_n} \left(\text{sgn}(f) \cdot a_{r,f(r)} \cdot \prod_{\substack{i \in [n] \\ i \neq r}} a_{i,f(i)} \right) + && // \text{since } c_{i,f(r)} = a_{i,f(r)}, \text{ for } i \in [n], i \neq r // \\
&\quad + \lambda_2 \cdot \sum_{f \in S_n} \left(\text{sgn}(f) \cdot b_{r,f(r)} \cdot \prod_{\substack{i \in [n] \\ i \neq r}} b_{i,f(i)} \right) && \text{and } c_{i,f(r)} = b_{i,f(r)}, \text{ for } i \in [n], i \neq r // \\
&= \lambda_1 \cdot \det(A) + \lambda_2 \cdot \det(B). && // \text{by Equality (6.1)} //
\end{aligned}$$

3) By Equality (6.3), we can assume that $A = [a_{i,j}]$ is upper triangular. If $f \in S_n$ and $f \neq id_n$ then, by Lemma 6.1, $f(i') < i'$, for some $i' \in [n]$. Therefore, $a_{i',f(i')} = 0$ and we have

$$\prod_{i \in [n]} a_{i,f(i)} = 0, \quad \text{for any } f \in S_n, f \neq id_n. \quad (6.5)$$

Thus:

$$\begin{aligned}
\det(A) &= \operatorname{sgn}(id_n) \cdot \prod_{i \in [n]} a_{i, id_n(i)} + \sum_{\substack{f \in S_n \\ f \neq id_n}} \left(\operatorname{sgn}(f) \cdot \prod_{i \in [n]} a_{i, f(i)} \right) \quad // \text{by Equality (6.1)} // \\
&= \prod_{i \in [n]} a_{i, i} + \sum_{\substack{f \in S_n \\ f \neq id_n}} \left(\operatorname{sgn}(f) \cdot \prod_{i \in [n]} a_{i, f(i)} \right) \quad // \text{since } id_n(i) = i \text{ and } \operatorname{sgn}(id_n) = 1 // \\
&= \prod_{i \in [n]} a_{i, i} + \sum_{\substack{f \in S_n \\ f \neq id_n}} (\operatorname{sgn}(f) \cdot 0) \quad // \text{by Equality (6.5)} // \\
&= \prod_{i \in [n]} a_{i, i}.
\end{aligned}$$

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When dealing with determinants, the following notations are convenient. If $A \in \operatorname{Mat}_n(\mathbb{F})$, we write

$$A = \left[[A]_{(1)}, [A]_{(2)}, \dots, [A]_{(n)} \right] \quad \text{and} \quad \det(A) = \left| [A]_{(1)}, [A]_{(2)}, \dots, [A]_{(n)} \right|, \quad (6.6)$$

where, as usual, $[A]_{(i)}$ is the i -th row of A . Adopting this notation, the first term of Proposition 6.1 can be rewritten as:

$$\text{if } [A]_{(r)} = [A]_{(s)} \text{ then } \left| \dots, [A]_{(r)}, \dots, [A]_{(s)}, \dots \right| = 0, \quad (6.7)$$

(in this notation, we omitted all rows of A except rows r and s)

and the second term of Proposition 6.1 can be rewritten as:

$$\left| \dots, \lambda_1 [A]_{(r)} + \lambda_2 [A]_{(s)}, \dots \right| = \lambda_1 \left| \dots, [A]_{(r)}, \dots \right| + \lambda_2 \left| \dots, [A]_{(s)}, \dots \right|. \quad (6.8)$$

Corollary 6.1. *If $A \in \operatorname{Mat}_n(\mathbb{F})$ has a zero row then $\det(A) = 0$.*

Proof. Suppose that $[A]_{(i)} = 0_{1,n}$, for some $i \in [n]$, then we have

$$\begin{aligned}
\left| \dots, [A]_{(i)}, \dots \right| &= \left| \dots, 0_{1,n}, \dots \right| \\
&= \left| \dots, 0_{1,n} + 0_{1,n}, \dots \right| \\
&\stackrel{(6.8)}{=} \left| \dots, 0_{1,n}, \dots \right| + \left| \dots, 0_{1,n}, \dots \right|.
\end{aligned}$$

That is, $\det(A) = \det(A) + \det(A)$, therefore $\det(A) = 0$.

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